

Unit 10, Lesson 8: Solving Real-World Problems with Two-Step Equations

Benchmark

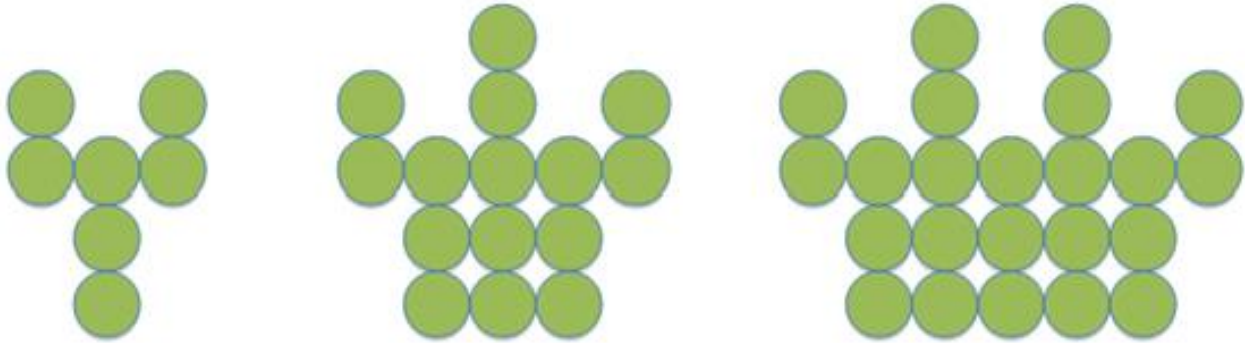
MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.

Learning Targets

- ☐ I can solve two-step equations within a real-world context.
- ☐ I can relate the solution of an equation representing a real-world context to the context.

10.8.1 Warm-Up: Number Patterns

A pattern is shown.

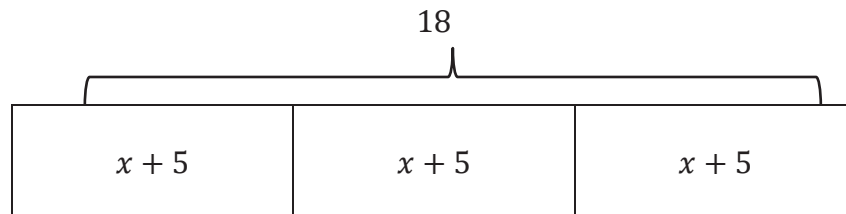


1. Describe how the pattern is changing in each step.
2. How many circles will be in the next step?

10.8.2 Collaboration: Matching Equations to Diagrams and Contexts

Equations can be represented using bar diagrams.

1. A bar diagram is shown.



- a. Select all the equations that match the diagram.
- ☐ $x + 5 = 18$
 - ☐ $18 \div 3 = x + 5$
 - ☐ $3(x + 5) = 18$
 - ☐ $x + 5 = \frac{1}{3} \cdot 18$
 - ☐ $3x + 5 = 18$
- b. Compare your selections with your partner. Resolve any differences if necessary.
- c. With your partner, choose one of the equations that can be modeled by the bar diagram and explain why it matches the diagram.
- d. Explain why $x + 5 = 18$ does not match the bar diagram.

Equations can be used to represent real-world contexts.

2. Three equations are shown.

Equation A	Equation B	Equation C
$\frac{1}{4} + 4x = 8$	$4 + \frac{1}{4}x = 8$	$8x + \frac{1}{4} = 4$

- a. Work with your partner to determine which equation matches each context. Then, write each equation in the appropriate row.

Context	Equation
A stack of nested paper cups is 8 inches tall. The first cup is 4 inches tall and each of the rest of the cups in the stack adds $\frac{1}{4}$ inch to the height of the stack.	
A baker uses 4 cups of flour. She uses $\frac{1}{4}$ cup to cover the counter with flour so the dough doesn't stick, and the rest to make 8 identical loaves of bread.	
Elena has an 8-foot piece of ribbon. She cuts off a piece that is $\frac{1}{4}$ of a foot long and cuts the remainder into four pieces of equal length.	

- b. With your partner, discuss what the variable x represents in each of the equations. Then, write the meaning of x in each equation based on the context it matches.

Equation	Meaning of x
A: $\frac{1}{4} + 4x = 8$	
B: $4 + \frac{1}{4}x = 8$	
C: $8x + \frac{1}{4} = 4$	

10.8.3 Guided Instruction: Relating Equations to Their Contexts

Olivia had her birthday party at a trampoline park. It cost \$48.50 to reserve a private jumping area for their group and \$15.72 per person for food, drinks, and unlimited jump time. Olivia's parents paid \$315.74 for Olivia and her friends to attend the party.

1. What was the total amount of money spent on the party?
2. One of the costs depends on the number of people attending the party, while the other does not.
 - a. Which cost depends on the number of people who attend the party?
 - b. Which cost is the same regardless of how many people attend the party?

The total cost for Olivia's birthday party can be represented by the equation $315.74 = 15.72n + 48.50$.

3. Work with your partner to answer the following questions.
 - a. What does n represent in the equation?
 - b. What does the variable term, $15.72n$, represent in the equation?

- c. What does the constant term, 48.50, represent in the equation?
4. Solve the equation $315.74 = 15.72n + 48.50$.
5. If Olivia was included as one of the people attending the party, determine how many of her friends attended.

10.8.4 Guided Practice: Relating Equations to Their Contexts

1. Franco is saving money to take a trip to Cocoa Beach, Florida with his friends. Every week, he sets aside the same portion of his paycheck after taking out the \$325 he needs for his weekly expenses. After 11 weeks, Franco has \$489.94 saved for his trip.

- a. An equation can be created relating these quantities as described below.

$$\left(\begin{array}{c} \text{Number of} \\ \text{weeks} \end{array} \right) \left(\begin{array}{c} \text{Amount of} \\ \text{weekly} \\ \text{paycheck} \end{array} - \begin{array}{c} \text{Amount of} \\ \text{weekly} \\ \text{expenses} \end{array} \right) = \begin{array}{c} \text{Total} \\ \text{Saved} \end{array}$$

Which of these quantities is unknown?

- b. Write the equation that represents this situation using the variable p for the unknown quantity.
- c. What does $(p - 325)$ represent in the equation?
- d. Solve the equation.
- e. Explain what the solution means in this context.

Work with your partner to complete the following.

- | Equation A | Equation B |
|-----------------------|---------------------|
| $46.92 = 6(y - 1.75)$ | $6y - 1.75 = 46.92$ |

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10.8.6 Wrap-Up: Reflection

1. Circle the value on the emoji scale that best represents your understanding of today's lesson.



2. Briefly explain your choice.

